

LCAP — Low-Code Application Platform

Designing a configurable platform to build, manage and automate enterprise applications

Role: UI/UX Designer

Project: LCAP — Low-Code Application Platform

Product Type: Enterprise SaaS / Low-Code Platform

Focus Areas: Product Design · UX Architecture · Form Builder · Report Builder · Workflow Builder · Data Management · Application Management

Overview

LCAP is a low-code application platform designed to help organisations create and manage business applications without requiring extensive development effort.

The platform brings multiple capabilities into one environment — from creating applications and defining data structures to building forms, generating reports, configuring workflows and setting up notifications.

The interface shown in the project demonstrates experiences such as application management, schema configuration, form creation, report creation, workflow configuration and notification setup.

The design goal

Create a scalable enterprise experience that allows users to configure business applications through a structured, visual interface rather than relying heavily on technical development.

The Challenge

Enterprise application builders can become complicated very quickly.

Users may need to work with:

- Applications
- Tables and fields
- Forms
- Reports
- Relationships
- Workflows
- Roles
- Notifications

- Permissions
- Records

LCAP needed to bring these capabilities together while maintaining a clear mental model for users.

The challenge was therefore to design an experience where users could progressively configure an application without feeling overwhelmed by the number of available capabilities.

The UX Approach

I structured the experience around the lifecycle of an application.

01 — Create

Users start by creating an application and defining its purpose.

For example, an application can be created with a name, alias and description.

02 — Structure

Users define the application's underlying data structure through tables, fields and relationships.

The platform includes concepts such as **Record of approvals, Inspection data, Environment impact, Construction progress, Financial data and Safety records.**

03 — Build

Once the data structure is established, users can create forms and configure the fields required to capture information.

The Form Builder provides content components such as text and text areas, along with form components including email, location, phone, radio buttons, checklists, checkboxes and select controls.

04 — Automate

Workflows allow teams to define steps, roles, preconditions, postconditions and outcomes.

This creates a foundation for automating business processes rather than treating every application as a collection of static forms.

05 — Communicate

Notifications can be configured through channels such as SMS and email, with dynamic fields that can be inserted into messages.

For example, the notification interface supports selecting an object and field when creating communication content.

06 — Analyse

The Report Builder allows users to construct reports using fields and configure grouping, filtering and sorting.

This enables users to turn application data into structured business reports.

Information Architecture

The primary application navigation is organised around the core building blocks of the platform:

Overview

- Applications
- Schema
- Forms
- Reports
- Dashboard
- Workflow
- Notification
- Settings

This structure creates a predictable relationship between the different stages of application configuration.

Why this matters

Instead of presenting every configuration capability simultaneously, the navigation separates major tasks into dedicated areas.

This helps users understand:

What am I building?

Application

What data does it use?

Schema / Tables / Fields

How do users enter information?

Forms

How is the information analysed?

Reports

How does the process operate?

Workflow

How are users notified?

Notifications

Application Management

The application overview provides users with multiple ways to scan and manage applications.

The interface supports:

- Grid View
- List View
- Filtering
- Sorting
- Search
- Application status
- Updated date
- Updated by

Applications can have statuses such as **Published** and **Deployed**.

This creates a familiar management pattern for enterprise users who need to work across multiple applications.

Schema & Data Management

A core part of the platform is the ability to define how application data is structured.

The table experience allows users to create and manage tables such as:

- Record of approvals
- Inspection data
- Environment impact
- Construction progress
- Financial data
- Safety records

Users can also navigate between **Tables** and **Fields**, providing a clear progression from high-level data structures to individual attributes.

Relationships

The schema experience also introduces relationships between data entities.

For example:

Record of approvals → Building codes

Users can define the relationship type and whether the relationship is mandatory.

Form Builder

The Form Builder is one of the most visual parts of the platform.

Users can construct forms using configurable components rather than manually designing every field.

Components include

Content

- Text
- Text Area

Form

- Email
- Location
- Phone
- Radio
- Checklist
- Checkbox
- Select

Media

- Image
- Video
- Upload

Configuration

Selecting a component exposes properties such as:

- Table
- Field

- Label
- Description
- Row configuration
- Default value

This establishes a component → property configuration pattern that can scale as more components are introduced.

Report Builder

The Report Builder transforms application data into configurable reports.

Users can select available fields and place them into the report canvas.

The experience also supports:

Grouping → Filtering → Sorting

This creates a simple configuration model for users who need to customise how information is presented without writing queries or code.

Example:

Compliance Audit Report

Data sources can include:

- Record of approvals
- Inspection Data
- Environment Impact

The report can then be configured using the required fields.

Workflow Builder

The workflow experience is designed around a step-based process.

A workflow can contain:

Pre-condition → Step → Role → Post-condition → Outcome → Notification

Users can add additional steps and assign roles to individual stages.

This approach helps transform complex business logic into a visual sequence.

Example

Step 1

Pre-condition

↓

Step 2

Assigned role

↓

Step 3

Post-condition

↓

Outcome

↓

Notification

The visual sequencing makes the relationship between process stages easier to understand.

Notification Builder

The notification system supports multiple communication channels.

SMS

Users can select an SMS gateway and construct the message using dynamic application data.

Email

Users can configure:

- Subject
- Message
- Object
- Field

Dynamic placeholders such as:

{first_name}

{company}

{industry}

allow application data to be inserted into communications.

This provides a reusable notification model rather than requiring users to manually create every message.

Design System Thinking

Because LCAP contains multiple builders and configuration-heavy interfaces, consistency becomes critical.

The experience relies on repeated patterns such as:

Navigation

Persistent navigation gives users access to the platform's primary modules.

Lists

Applications, forms, reports and records use repeatable list structures.

Builders

Form Builder and Report Builder follow a component/configuration model.

Progressive configuration

Users move from:

Object → Component → Property → Configuration

This pattern can be reused throughout the platform.

Key UX Principles

01 — Progressive Disclosure

Expose complexity when users need it instead of presenting every configuration option at once.

02 — Consistent Patterns

Use the same interaction model across Forms, Reports, Tables and Workflows.

03 — Clear Hierarchy

Separate application-level decisions from field-level configuration.

04 — Visual Configuration

Use builders and structured controls wherever possible instead of requiring technical input.

05 — Scalability

Design patterns should work for both a small application and an organisation managing many applications.

My Design Contribution

As a UI/UX Designer, the work can be positioned around:

- Information architecture
 - Enterprise UX architecture
 - Application management experiences
 - Form Builder UX
 - Report Builder UX
 - Workflow Builder UX
 - Schema and data management
 - Component-based interaction design
 - Configuration patterns
 - Responsive and scalable UI patterns
 - Design consistency across complex workflows
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Outcome

The resulting experience brings multiple stages of application development into a single platform:

Create → Structure → Build → Automate → Notify → Analyse

Instead of treating application development, data management, form creation, reporting and workflow automation as separate tools, LCAP connects them through a common configuration experience.

The PDF demonstrates this end-to-end ecosystem through application creation, table and field management, form building, report creation, workflow configuration and notification setup.

What I Learned

Designing an enterprise low-code platform reinforced an important principle:

The more powerful a product becomes, the more important its information architecture becomes.

The challenge isn't simply giving users more capabilities.

It is helping them understand **where each capability belongs, what it affects and what they should do next.**

For LCAP, the design focus therefore shifted from individual screens to the larger system connecting applications, data, forms, reports, workflows and notifications.

Case Study Summary

LCAP

Low-Code Application Platform

Designed for: Enterprise application configuration

Core UX:

Application Management · Schema · Forms · Reports · Workflow · Notifications

Design approach:

Structured · Modular · Scalable · Configuration-driven

Key takeaway:

Making complex enterprise configuration feel simple through clear information architecture and reusable interaction patterns.